

ICT2217 Network Security
Lab Exercise 3.4: AAA Part 3 - Accounting

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Configure accounting using centralised TACACS+ and RADIUS servers

Required Hardware

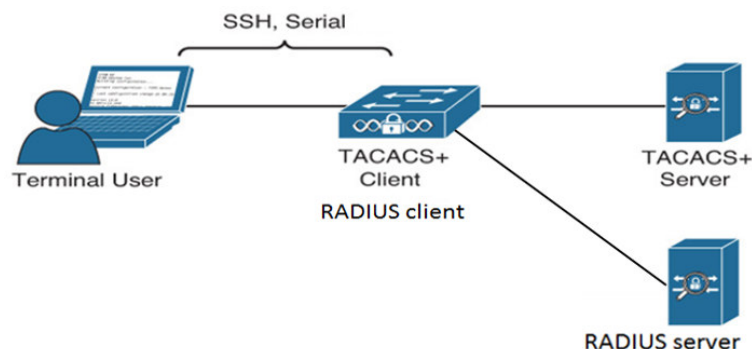
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-RJ45 console cables
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive
- TACACS.net 2.1.2 Demo version for Windows
<https://www.softpedia.com/get/Internet/Servers/Other-Servers/TACACS-net.shtml>
- FreeRADIUS 3.2.3 for Kali Linux

PART 1: Configuring Accounting using TACACS+ and RADIUS Servers

- 1.1 Continue with the same network topology in Lab Exercise 3.3.



- 1.2 Configure accounting for 'exec' as shown in Lab 3.4 Notes pg 3.
- 1.3 In addition, configure accounting for 'commands' as shown in Lab 3.4 Notes pg 8.
- 1.4 When done, enter 'exit' until you are disconnected.

- 1.5 Now, login again using SSH.
- 1.6 Go to C:\ProgramData\TACACS.net\Logs folder and open the log file. Are you able to see the start exec log at TACACS+ server similar as that shown on Lab 3.4 Notes pg 7?
- 1.7 Try running some privilege level 15 commands. Are you able to see the commands being logged at TACACS+ server similar as that shown on Lab 3.4 Notes pg 8?
- 1.8 Enter 'exit' until you are disconnected. Are you able to see the stop exec being logged at TACACS+ server?
- 1.9 Next, simulate TACACS+ server is down by disconnecting it from the switch.
- 1.10 Login again using SSH.
- 1.11 Go to /var/log/freeradius directory and open the log file. Are you able to see the start exec being logged at RADIUS server similar as that shown in Lab 3.4 Notes pg 5?
- 1.12 Finally, enter 'exit' until you are disconnected. Are you able to see the stop exec being logged at the RADIUS server?